

X-Ray Diffraction of Infrared Quantum Bragg Mirror Detector Structures Designed by Genetic Algorithm

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Abstract—This work presents the growth and structural characterization of Infrared Quantum Bragg Mirror Detectors (QBMDs) heterostructures based on InGaAs/InAlAs heterostructures lattice matched to InP. Two distinct superlattice structures were previously designed using computational simulations orchestrated by a Genetic Algorithm to optimize key figures of merit, specifically oscillator strength and photocurrent peak for an optical transition around 300 meV. The samples were grown by Metalorganic Vapor Phase Epitaxy and subsequently analyzed using High-Resolution X-Ray Diffractometry to verify crystalline quality, layer thickness, and composition. The experimental results demonstrate high structural precision, with layer thickness deviations generally remaining below one atomic layer (approximately 2.7 Å) and compositions at the nominal designed values.

Index Terms— Quantum Bragg Mirror Detectors, Infrared Photodetectors, X-Ray Diffractometry, Metalorganic Vapor Phase Epitaxy, Artificial Intelligence.

I. INTRODUCTION

Photodetectors are devices that convert light energy into an electrical current or potential. They are designed following principles of condensed matter physics and rely on quantum confinement effects for photon detection [1]. The research and development of infrared photodetectors has been strongly driven by strategic applications, such as defense, remote sensing, satellite imaging, thermal imaging, and motion detection [2], [3]. Our research group has previously demonstrated direct applications of such devices, such as measuring the protein percentages in milk with a quantum dot infrared photodetector (QDIP) [4], and proposing gas concentration monitoring system using a Quantum Bragg Mirror Detector (QBMD) [5].

A superlattice (SL) consists of a series of quantum wells (QWs) coupled by resonant tunneling. An important parameter for observing quantum effects in these structures is the phase-coherence length. Since this effect depends heavily on the bulk and interface quality of the crystals, as well as temperature and effective mass, precise control over the material growth is essential [6], [7]. By manipulating the composition and dimensions of the heterostructure, it is possible to modulate the potential wells, thereby selecting the absorbed photon energy and optimizing relevant figures of merit.

An electronic Bragg mirror is an analogue of a Fabry-Pérot cavity for light [8]. The Bragg condition imposes that the layers thicknesses should match a quarter-wavelength at the electronic transmission resonance of the quantum well.

However, this restriction is not mandatory to confine a state above the barrier [9], [10]. By relaxing this condition, one can create bound and quasi-bound state in the continuum that can be used for optoelectronic devices [11].

However, finding the optimal configuration for these multi-layered structures is a complex multi-parameter problem. To address this, we employed a Genetic Algorithm (GA) to orchestrate computational simulations and design two unique QBMD structures targeting maximized oscillator strength and/or photocurrent peak. To validate the feasibility of these computer-designed structures, the samples were grown by Metalorganic Vapor Phase Epitaxy (MOVPE) and characterized by High-Resolution X-Ray Diffractometry (HRXRD) to ensure the structural precision required for obtaining the bound and quasi-bound states in the continuum in the QBMDs.

II. METHODOLOGY

A. Structure design and optimizations

The photodetector structures were designed based on In_{0.53}Ga_{0.47}As quantum wells, with a bandgap energy (E_g) of 0.74 eV, and In_{0.52}Al_{0.48}As quantum barriers ($E_g = 1.45$ eV), materials that can be grown lattice-matched to InP. The conduction band offset energy of the In_{0.53}Ga_{0.47}As/In_{0.52}Al_{0.48}As heterostructure is 503 meV. The *e*-mulate software [12] was used to obtain the electronic band structure and figures of merit of heterostructures. This software is based on transfer matrix method to calculate the electronic wavefunctions and their energies. As the energies involved are comparable to the bandgap of the material, a non-parabolic approximation is used [13]. The correction for these effects is incorporated into the effective mass of the electron, which becomes dependent on the electron energy [14]. We used the oscillator strength (OS) and photocurrent spectrum (PC) as the figures of merit to be maximized during the optimizations.

Fig. 1(a) presents the electronic conduction band structure of the reference sample (NR1005), which served as the starting point for optimizations. This structure consists of a 2.5 nm thick main quantum well (MQW) surrounded by two superlattices. The left superlattice is composed of five lateral quantum wells (LQWs) with a thickness of 2.0 nm and lateral quantum barriers (LQBs) of 7.0 nm. The right superlattice comprises a single RQW and RQB with identical thicknesses to those on the left. These surrounding superlattices were

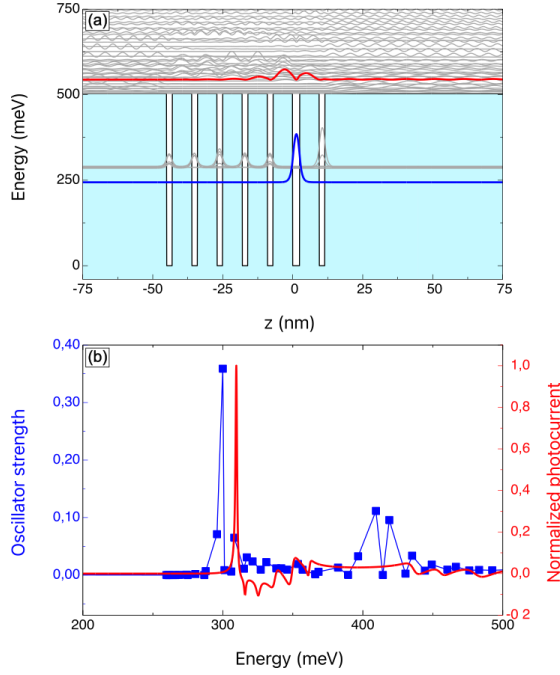


Fig. 1: (a) Energy diagram of the reference sample (NR1005) with 5 quantum wells on the left side and one quantum well on the right side of the main quantum well. Wavefunctions are shown in grey, with the ground state highlighted in blue and the quasi-bound state with highest oscillator strength highlighted in red. (b) Oscillator Strength and Photocurrent of the reference sample.

tuned to create a quasi-bound state in the continuum (red line) with an energy of approximately 540 meV. Fig. 1(b) displays two figures of merit: the oscillator strength, used to estimate optical transitions, and the photovoltaic photocurrent spectrum, which represents the current generated by photon absorption. In both spectra, a main peak is observed near 300 meV. This peak corresponds to the optical transition between the ground state (blue line) and the quasi-BIC (red line). The structure shown represents a single period of the heterostructure, which was repeated 20 times during sample growth to enhance absorption.

Considering the heterostructure shown in Fig. 1 as a reference, two different optimization strategies were investigated to enhance its figures of merit for the optical transition near 300 meV. The first approach examined how the number of quantum wells and barriers in each Bragg mirror surrounding the MQW affects the OS and PC. In the second approach, the GA was employed to identify other heterostructures that outperform the reference structure. The objective function for the GA optimizations was designed to maximize both the oscillator strength and the photocurrent peak for an optical transition around (300 ± 10) meV. Further details regarding the optimization process and the parameters used can be found in [14]. This specific energy aligns with the optical absorption band of carbon dioxide (CO₂) [9], enabling direct applications in gas sensing for environmental monitoring and leak detection.

During the GA optimization process, up to seven parameters were varied to refine the structures. These parameters included: the number and thickness of the lateral quantum wells (LQWs) and the thickness of the quantum barriers (LQBs) on the left side; the thickness of the main quantum well (MQW) in the center; and the number and thickness of the quantum wells (RQWs) along with the thickness of the barriers (RQBs) on the right side. Table 1 summarizes the

Table 1: Structures of the designed samples.

Sample	# of LQW	LQW (nm)	LQB (nm)	MQW (nm)	# of RQW	RQB (nm)	RQW (nm)
NR1005	5	2.0	7.0	2.5	1	7.0	2.0
NR1006	6	2.0	7.0	2.5	4	7.0	2.0
NR1007	7	2.0	7.0	2.5	4	7.0	2.0
NR1008	7	1.1	7.4	2.5	4	7.4	1.1
NR1009	7	2.0	7.1	2.7	6	7.8	1.8
NR1010	8	2.0	7.0	2.5	4	7.0	2.0

structures selected for growth: the reference sample (NR1005), the three samples from the first optimization approach (NR1006, NR1007, and NR1010), and the two GA-optimized samples (NR1008 and NR1009).

Table 2 presents the OS and PC results from simulations, including a relative comparison between the reference sample (NR1005) and the simulated figures of merit of the optimized samples. This comparison demonstrates that all samples are expected to present stronger OS, with gains ranging from 24 to 63%. Photocurrent amplitude is greatly increased, showing a theoretical improvement of more than two orders of magnitude. Although experimental measurements have not yet been performed, these promising results indicate that the simulated structures could be significantly better photo-detectors compared to the reference sample.

B. Sample fabrication

The QBMD's heterostructures were fabricated with alternating quantum wells of In_{0.53}Ga_{0.47}As and In_{0.52}Al_{0.48}As. The samples were grown by MOCVD, lattice-matched to 2" monocrystalline undoped InP wafers, at 100 mbar and 650 °C. Trimethyl-indium, trimethyl-gallium, trimethyl-aluminum, arsine and phosphine were used as precursor gases for In, Ga, Al, As and P, respectively.

Epitaxial growth starts with the deposition of a 300 nm undoped InP buffer layer, then an 800 nm Si-doped InGaAs contact layer is deposited at a high growth rate. A 30 nm InAlAs spacer layer is deposited at a high growth rate, before the start of the deposition of 20 periods of the superlattice's fundamental structure, as defined in Table 1 for each sample. The main QW is *n*-doped with Si using a flux of silane (SiH₄)

Table 2: Simulation results for the oscillator strength and photocurrent of all samples, including the gain, relative to the reference sample's figures of merit.

Sample	Oscillator Strength			Photocurrent	
	OS	Relative Gain (%)	Peak Energy (meV)	Normalized PC	Peak Energy (meV)
NR1005	0.359	0%	299.9	1.0	309.6
NR1006	0.449	25%	298.5	39.4	308.0
NR1007	0.449	25%	298.5	49.9	308.2
NR1008	0.586	63%	303.2	134.8	312.0
NR1009	0.446	24%	296.2	381.3	315.9
NR1010	0.449	25%	298.5	53.3	308.2

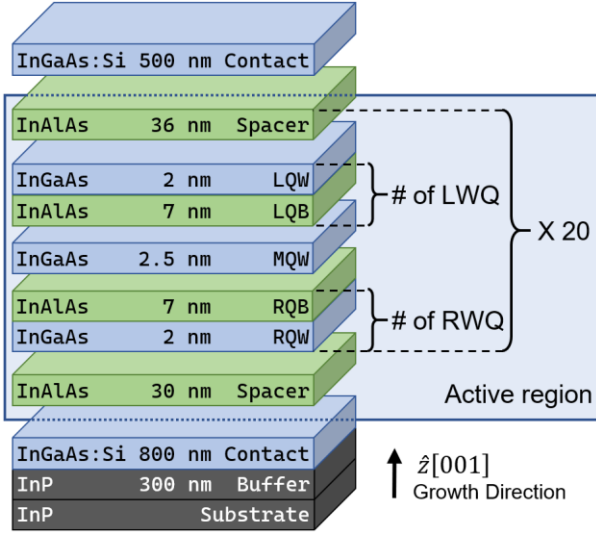


Fig. 2: Base structure of the samples. The growth direction shows the crystalline orientation of the quantum wells.

to achieve 2×10^{18} electrons per cm^{-3} . These layers are deposited at a low growth rate of $1.1 \text{ \AA/s}$, in order to obtain high crystalline quality, fine thickness control, and high layer uniformity. Between each SL period a 36 nm InAlAs separator layer is fabricated. The topmost contact is a contact layer with the same composition as the first one with a thickness of 500 nm . A schematic of the reference structure is shown in Fig. 2.

C. High-Resolution X-Ray Diffractometry

Following the growth of all samples, HRXRD analysis was performed, aiming to measure the thickness and composition of the deposited layers. The measurements were performed on a Bruker D8 Discover diffractometer and the data was analyzed with the commercial Bruker Leptos software. A typical analysis of HRXRD spectrum is performed through curve fitting to the simulated spectrum for the nominal designed structure. A good adjustment between curves allows a precise estimation of the thickness and composition of each layer.

III. RESULTS AND DISCUSSION

Fig. 3 presents the diffractogram of the reference sample, with the typical interference fringes of superlattices. The good match between measured spectra and the fitted

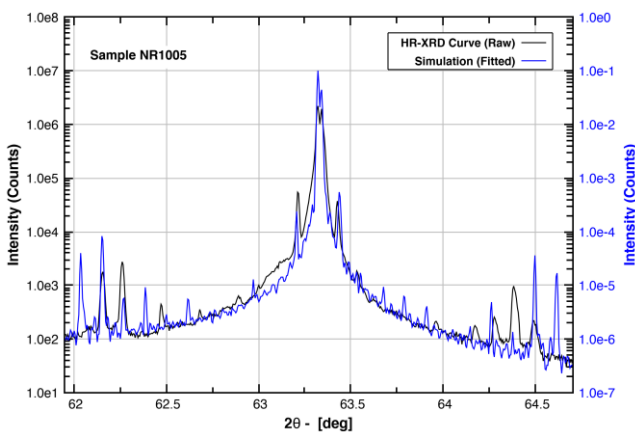


Fig. 3: Measured (Raw) and simulated diffractograms of the reference sample.

Table 3: Thickness and composition analysis for sample NR1005.

NR1005	Thickness (nm)			Composition (% In)	
Layer	Nominal	Measured	Error	Nominal	Measured
LQW	2	2.03 ± 0.06	1.3%	53	53.22 ± 0.04
LQB	7	6.51 ± 0.22	-6.9%	52	52.19 ± 0.08
MQW	2.5	2.48	-0.7%	53	53.16
RQB	7	6.94	-0.9%	52	52.36
RQW	2	1.93	-3.3%	53	53.23

simulation indicates that the sample's crystalline quality is good, and that the layer's thicknesses and composition are close to the nominal values from the design. The overlap between the interference peaks of the substrate and the epitaxial layers grown on it, which differ by a few arc-seconds, shows an excellent lattice-matched structure.

Table 3 presents the nominal and measured values of thickness and composition for each layer group in the reference sample. The average thickness of almost all layers deviates less than $1 \text{ \AA}$ from the nominal value, meaning that every layer's composition is close to the nominal value. The thickness of the left quantum barriers is the one that deviated the most from the nominal value – approximately 1.7 monolayers (ML) thinner than designed – demonstrating a precise growth control. The indium content is slightly higher than nominal for all the samples, showing that a very small offset occurred during the calibration of growth conditions, but still under experimental tolerances expected for the growth.

For comparison, the diffractogram of NR1009 is shown in Fig. 4, revealing more satellite peaks. This is explained by the fact that this sample contains more QWs on both sides of the MQW, creating more interference fringes. Table 4 presents the HRXRD analysis of thickness and composition of sample NR1009. The average thickness and its standard deviation were calculated for the QWs and QBs only when more than one pair of layers was present in at least one side of the structure. This limitation is due to the simulation and fitting of HRXRD that considers the 20x repetitions in the active region as identical. Growth quality and consistency are demonstrated by the very good match between designed and measured results.

For all studied samples, layer thicknesses are within fabrication tolerance, indicating that the structures present very consistent periodicity, essential for obtaining the desired

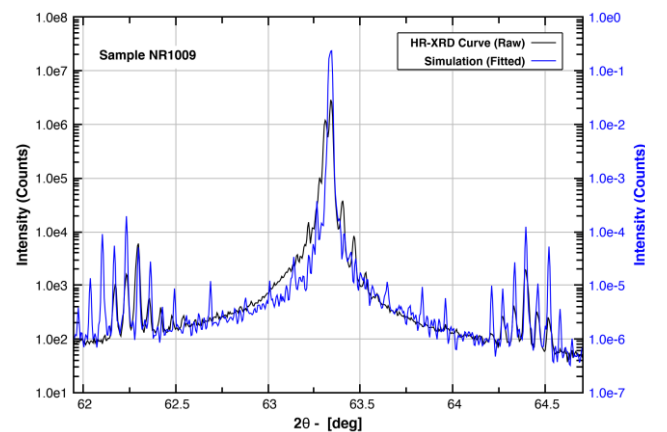


Fig. 4: Measured (Raw) and simulated diffractograms of sample NR1009.

Table 4: Thickness and composition analysis for sample NR1009

NR1009 Layer	Thickness (nm)			Composition (% In)	
	Nominal	Measured	Error	Nominal	Measured
LQW	2	1.93±0.07	-3.3%	53	53.21±0.03
LQB	7.1	7.16±0.36	0.9%	52	52.29±0.05
MQW	2.7	2.71	0.4%	53	53.17
RQB	7.8	8.21±0.30	5.3%	52	52.24±0.04
RQW	1.8	1.68±0.17	-6.5%	53	53.21±0.02

QBMD optoelectronic properties. Most standard deviations are below one atomic layer thickness (approximately 2.7 Å for InP), highlighting the extremely well-controlled epitaxial growth process.

IV. CONCLUSION

We have successfully demonstrated the fabrication and structural validation of InGaAs/InAlAs QBMDs optimized via artificial intelligence. Computational optimization predicts remarkable performance enhancements, including a 63% increase in oscillator strength and a photocurrent gain of more than two orders of magnitude, compared to the reference device. High-Resolution X-Ray Diffractometry measurements confirmed that the MOVPE growth process achieved exceptional structural fidelity, with layer thickness and composition matching the nominal designs within growth tolerances. Specifically, thickness deviations were maintained below one monolayer, ensuring the periodicity required for the observation of quantum confinement in the continuum. Future efforts will focus on the optoelectronic characterization of these devices to experimentally validate the predicted oscillator strength and photocurrent efficiency.

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